Université Paris Sorbonne Nord
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Master 1.
Distributed Java Programming
TP: Protocol Handlers

MyFTP

In this exercise, we shall use the classes

- `URL` : `https://docs.oracle.com/javase/7/docs/api/java/net/URL.html`
- `URLConnection` : `https://docs.oracle.com/javase/7/docs/api/java/net/URLConnection.html`
- `URLStreamHandler` : `https://docs.oracle.com/javase/7/docs/api/java/net/URLStreamHandler.html`

to implement a simple FTP protocol. In the file `DateService.java` you will find an example of the use of these classes. Additional examples and information can be found at the Java API documentation. See also, e.g., `https://www.oreilly.com/library/view/learning-java/1565927184/apas02.html`. Before start coding, read carefully the API documentation to understand the purpose of each of the above classes.

The idea is to implement a simple protocol that allows us to download files from the server by using URLs of the form:

`myftp://localhost:2222?myfile.txt`

Here some ideas:

- The class `URL` offers the needed methods to retrieve the port (2222), the host (localhost) and the queries (?myfile.txt) from a given URL.
- The method `connect` in the class extending `URLConnection` must establish a connection, using a Socket, to the server. You can also use this method to send the name of the requested file to the server.
- Modify the method `getContent` to retrieve the file and store it locally.
- Implement a server (ServerSocket) to listen to the requests from the clients. You should use an ExecutorService to handle the different connections from the clients.

- **(Extra)** Use the queries from the URL to implement different services. For instance, one may ask to the server the list of all the available files, the size of a file or download all the files that match a given pattern.